

AUBURN-LEWISTON, ME, USA

WMO: 726184

Lat: 44.050N

Lon: 70.283W

Elev: 288

StdP: 14.54

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94709

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	0.1	-20.4	1.8	1.1	-14.9	2.5	3.8	24.6	19.4	21.8	19.9	3.8	10	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	87.8	71.5	83.8	70.0	81.2	67.7	74.2	83.3	72.1	80.4	70.3	77.4	9.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.6	118.1	79.6	69.8	110.9	77.1	67.7	102.7	74.9	37.8	83.2	36.0	80.0	34.3	77.1	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.8	(b) 18.4	(c) 16.3	(e) -14.4	(f) 92.3	(g) 6.9	(h) 2.5	(i) -19.3	(j) 94.1	(k) -23.3	(l) 95.5	(m) -27.2	(n) 96.9	(o) -32.2	(p) 98.7		
(4) 20.8	(5) 18.4	(6) 16.3	(7) -14.5	(8) 77.2	(9) 6.0	(10) 1.9	(11) -18.8	(12) 78.6	(13) -22.3	(14) 79.7	(15) -25.7	(16) 80.8	(17) -30.1	(18) 82.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.3	20.0	23.0	31.6	43.1	53.7	63.1	68.7	67.1	59.9	47.9	37.0	26.7		
	DBStd	18.61	10.98	10.32	9.48	7.28	6.94	6.82	5.21	5.11	7.05	7.49	8.43	9.64		
	HDD50	3786	930	757	575	230	36	1	0	0	5	131	398	723		
	HDD65	7539	1395	1176	1036	656	359	116	22	35	185	531	840	1188		
	CDD50	2056	0	0	4	25	149	394	580	530	301	66	7	0		
	CDD65	336	0	0	0	0	8	59	137	100	31	1	0	0		
	CDH74	3170	0	0	8	21	158	588	1205	882	290	18	0	0		
	CDH80	852	0	0	1	4	45	162	330	231	78	1	0	0		
	WSAvg	5.9	6.3	6.5	7.3	7.1	6.4	5.4	4.9	4.6	5.0	5.8	5.9	6.0		
	PrecAvg	44.7	3.4	3.3	4.0	4.1	3.7	3.5	3.3	3.0	3.4	4.1	4.6	4.3		
(15) Precipitation	PrecMax	65.9	10.4	7.8	12.0	8.7	11.4	11.1	7.3	10.8	13.2	13.1	11.6	12.6		
	PrecMin	29.6	0.3	0.1	0.7	0.5	0.6	0.7	0.6	0.2	0.6	0.7	1.2	1.2		
	PrecStd	8.0	1.9	1.5	2.0	1.7	2.2	2.0	1.5	1.8	2.0	2.4	2.1	2.4		
	PrecStd	8.0	1.9	1.5	2.0	1.7	2.2	2.0	1.5	1.8	2.0	2.4	2.1	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	51.8	66.4	76.9	86.1	90.0	90.7	90.4	88.2	76.7	63.8	54.9		
		MCWB	49.3	44.4	53.5	59.9	66.0	72.6	74.7	71.0	71.5	63.3	54.4	50.0		
	2%	DB	45.7	45.8	54.6	66.4	78.7	84.1	87.3	85.6	81.4	70.2	58.8	48.3		
		MCWB	41.9	39.6	44.7	53.1	61.7	70.1	72.4	70.3	68.0	59.7	52.9	44.8		
	5%	DB	39.3	41.3	48.3	61.3	72.5	80.9	83.6	81.9	76.7	64.4	54.8	44.9		
		MCWB	35.5	35.9	41.0	49.0	58.6	67.3	70.6	68.3	65.8	57.0	50.0	40.5		
	10%	DB	35.9	37.1	45.0	56.6	67.6	76.8	81.0	79.2	72.7	61.3	50.5	39.2		
		MCWB	32.2	32.9	38.3	45.8	56.1	64.0	68.5	66.5	63.4	54.4	45.6	35.0		
	0.4%	WB	50.6	46.1	54.9	61.4	68.1	75.3	77.0	76.2	73.6	66.4	59.6	52.0		
		MCDB	52.1	49.0	65.3	71.0	80.1	85.4	87.1	85.1	83.7	70.4	62.7	54.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.0	41.0	47.0	56.4	64.1	72.4	74.6	73.2	70.4	62.2	54.9	45.8		
		MCDB	45.6	45.2	51.6	64.9	73.4	81.7	83.4	82.0	77.1	66.8	57.2	47.6		
	5%	WB	36.7	37.0	41.8	51.2	60.6	69.7	72.5	70.9	68.0	59.1	50.5	40.2		
		MCDB	39.7	40.3	47.5	57.7	69.8	77.3	80.5	78.2	73.6	64.7	54.2	44.1		
	10%	WB	32.3	33.7	38.5	47.4	57.5	66.9	70.5	68.9	65.5	55.4	46.2	36.4		
		MCDB	35.1	37.2	43.6	55.4	65.6	73.8	77.6	75.7	70.6	60.2	49.3	39.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	18.6	17.7	19.9	20.6	20.3	20.3	21.5	21.3	19.7	16.6	16.5		
		MCDBR	19.8	21.0	23.9	29.0	30.0	26.3	24.6	25.2	26.2	25.9	21.8	20.9		
		MCWBR	17.7	16.4	16.1	17.1	16.0	14.1	11.9	12.8	14.8	17.1	17.2	17.6		
		MCDBR	19.8	19.2	22.0	25.4	25.4	22.9	20.6	21.4	21.3	22.3	18.8	19.8		
	5% WB	MCWBR	18.7	16.2	16.1	17.0	14.8	13.7	11.5	12.0	13.6	16.6	16.8	18.1		
		MCWBR	18.7	16.2	16.1	17.0	14.8	13.7	11.5	12.0	13.6	16.6	16.8	18.1		
(40) Clear-Sky Solar Irradiance	taub		0.276	0.293	0.326	0.351	0.380	0.432	0.445	0.412	0.372	0.332	0.315	0.285		
	taud		2.423	2.365	2.399	2.410	2.375	2.239	2.214	2.300	2.397	2.508	2.517	2.488		
	Ebn at Noon		272	286	290	290	283	267	262	267	270	269	256	259		
	Edh at Noon		23	29	33	35	37	43	44	39	33	26	22	20		
(44) All-Sky Solar Radiation	RadAvg		516	797	1165	1445	1614	1741	1809	1646	1334	844	547	407		
	RadStd		41	65	69	155	200	145	137	91	79	76	52	40		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	N/A

Nomenclature: See separate page